

$$\textcircled{1} \quad \textcircled{a} \quad P_{X,Y}(1,1) = P_X(1) P_Y(1). \rightarrow \textcircled{1}$$

$$\begin{aligned} P_{X,Y}(1,0) &= P_X(1) - P_{X,Y}(1,1). \\ &= P_X(1) - P_X(1) P_Y(1). \quad (\text{from } \textcircled{1}) \\ &= P_X(1) (1 - P_Y(1)) \\ &= P_X(1) P_Y(0). \end{aligned}$$

$$\begin{aligned} P_{X,Y}(0,1) &= P_Y(1) - P_{X,Y}(1,1). \\ &= P_Y(1) (1 - P_X(1)) \\ &= P_Y(1) P_X(0) \end{aligned}$$

$$\begin{aligned} P_{X,Y}(0,0) &= P_X(0) - P_{X,Y}(0,1). \\ &= P_X(0) - P_X(0) P_Y(1) \\ &= P_X(0) (1 - P_Y(1)) = P_X(0) P_Y(0). \end{aligned}$$

$\Rightarrow X, Y$ are independent

⑥

$$\text{Cov}(X, Y) = 0$$

$$\Rightarrow E[XY] = E[X]E[Y]$$

$$\begin{aligned} E[XY] &= P_{X,Y}(1,1) \times 1 \times 1 + P_{X,Y}(0,1) \times 0 \times 1 \\ &\quad + P_{X,Y}(1,0) \times 1 \times 0 \\ &\quad + P_{X,Y}(0,0) \times 0 \times 0 \\ &= P_{X,Y}(1,1) \end{aligned}$$

$$E[X] = P_X(1), \quad E[Y] = P_Y(1).$$

$$\Rightarrow P_{X,Y}(1,1) = P_X(1) P_Y(1).$$

$\Rightarrow X, Y$ are independent (from part (a))

② a) Let $Y_i = \begin{cases} 1 & \text{if box } i \text{ is empty} \\ 0 & \text{otherwise} \end{cases}$

$Y = Y_1 + Y_2 + \dots + Y_5 \quad := \# \text{ boxes that do not have any balls.}$

$$E[Y] = \sum_{i=1}^5 E[Y_i]$$

$$= \sum_{i=1}^5 P(Y_i = 1).$$

$$P(Y_i = 1) = (1 - p_i)^{10}.$$

$$\Rightarrow E[Y] = \sum_{i=1}^5 (1 - p_i)^{10}.$$

⑥ Let $Z_i = \begin{cases} 1 & \text{if box } i \text{ has exactly 1 ball} \\ 0 & \text{otherwise} \end{cases}$

$Z := \# \text{ boxes with exactly 1 ball.}$

$$E[Z] = \sum_{i=1}^5 P(Z_i = 1) = \sum_{i=1}^5 \binom{10}{1} p_i (1 - p_i)^9.$$

③ a) $X_i := \# \text{ balls in box } i$.

Each ball is placed in a box i w.p. p_i .

Successful trial if ball is placed in box i

Unsuccessful otherwise.

There are 10 balls / 10 trials

$$\Rightarrow X_i \sim \text{Binomial}(10, p_i)$$

$$P(X_i = x) = \binom{10}{x} p_i^x (1 - p_i)^{10-x}.$$

⑥ $X_i + X_j$

Method 1 : Using joint PMF

$$P_{X_i, X_j}(x, \hat{x}) = \binom{10}{x} \binom{10-\hat{x}}{\hat{x}} p_i^x p_j^{\hat{x}} (1 - p_i - p_j)^{10-x-\hat{x}}.$$

↓
Can't assume X_i, X_j are independent,

$$\text{Let } Y = X_i + X_j$$

$$P_Y(y) = \sum_{\substack{x=0 \\ \hat{x}=y-x}}^y P_{X_i, X_j}(x, \hat{x}).$$

$$= \sum_{x=0}^y \frac{10!}{(10-y)! x! (y-x)!} \frac{y!}{y!} p_i^x p_j^{y-x} (1-p_i-p_j)^{10-y}$$

$$= \frac{10!}{y! (10-y)!} (1-p_i-p_j)^y \sum_{x=0}^y \binom{y}{x} p_i^x p_j^{y-x}$$

$$= \binom{10}{y} (p_i + p_j)^y (1-p_i-p_j)^{10-y}.$$

Method 2 :

$$Y = X_i + X_j = \# \text{ balls in boxes } i \text{ or } j.$$

Successful trial if ball lands in box i or j .

Probability for this trial to be successful $= p_i + p_j$.

$$\Rightarrow Y \sim \text{Binomial}(10, p_i + p_j).$$

$$4. \quad E[X] = 1, \quad \text{Var}(X) = 5 \quad \Rightarrow \quad E[X^2] = \text{Var}(X) + (E[X])^2 \\ = 5 + 1 = 6.$$

$$E[(2+X)^2] = E[4 + 2X + X^2]$$

$$= 4 + 2 \times E[X] + E[X^2]$$

$$= 4 + 2 + 6 = 12.$$

$$\text{Var}(4+3X) = 9 \text{Var}(X) = 45.$$

X_1 : first outcome of dice
 X_2 : second outcome of dice
 }
 X_1, X_2 independant & identical

$$X = X_1 + X_2, \quad Y = X_1 - X_2$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

$$= E[(x_1 + x_2)(x_1 - x_2)] - (E[x_1 + x_2]E[x_1 - x_2])$$

$$= E[x_1^2 - x_2^2] - (E[x_1] + E[x_2])(E[x_1] - E[x_2])$$

$$= E[x_1^2] - E[x_2^2] - ((E[x_1])^2 - (E[x_2])^2)$$

$$= \text{Var}(x_1) - \text{Var}(x_2) = 0.$$

⑤

Let E_n be the expected # of steps needed in order to go from $(0,0)$ to $(n,0)$. (or from $(x,0)$ to $(x+n,0)$)

$$E_{10} = \frac{1}{2}(1 + E_9) + \frac{1}{2}(1 + E_{11}),$$

$$= 1 + \frac{1}{2}E_9 + \frac{1}{2}E_{11}.$$

1.5 marks

1.5 mark.



$$E_0 = 0.$$

$$\text{or } E_1 = \frac{1}{2}(1 + E_2) + \frac{1}{2}.$$

* There is no stopping condition to the left. So we won't be able to get closed form solution for E_{10} .